

EMILE HARMEL

Senior Software Engineer · TypeScript / Node.js / Next.js · Agentic AI systems

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§01 SUMMARY

GENERAL ARRANGEMENT

Architected and built a 264,000-line TypeScript platform in fourteen weeks, where AI agents write, illustrate, voice and film lessons for schools — and code, not the model, decides what ships.

Twenty years building web products, the last decade in TypeScript and React — enterprise platforms, national e-health, pan-European automotive. At MonstoryX I own the full stack: a Fastify API, three Node agent services, three Next.js apps, Postgres with row-level security, Cloud Run and Vercel. In pre-alpha with teachers. Built with a team of coding agents I orchestrate: owner agents per domain, separate critics, 36 ADRs.

§02 THE PLATFORM

MONSTORYX — WHAT IS BUILT AND RUNNING

SYS-01 A MULTI-AGENT VIDEO PIPELINE

Four agents turn a quest into a narrated film: brief → script → storyboard → keyframes (4 in parallel, ~17 s each) → vision QA → per-shot render → ffmpeg assembly. Each output is linted in code (~28 violation types) and retried once with the exact violations fed back. Every clip must prove it speaks its line — cross-correlation ≥ 0.95, lag ≤ 100 ms — or it re-renders. Failure, retry and refund are per slot; HMAC-signed callbacks retry with backoff and dedupe under a row lock.

STACK → Flue, Hono, Gemini, fal.ai (Nano Banana 2, Wan 3.0), Resemble, ffmpeg, Cloud Run

SYS-02 CHARACTER BIBLES AS EXECUTABLE CANON

Four characters' sheets, camera grammar, style register, story format and vendor recipes live in one package, compiled into agent prompts at build. Trait schemas are derived from the bible and hash-locked to the reference plates by tests. A vision judge grades each keyframe trait by trait on crops, colour as a hue band — calibrated from 5 false fails on 6 good frames to 3/3 defects caught, 0 false.

METHOD → canon as a package, derived schemas, calibrated vision judge · voice and persona graded by quest level

SYS-03 A CO-AUTHOR, QUEST IMAGES AND VOICE

9 agent workflows and 16 skills co-author quests in English and German, with tools built per request; invented words are dropped by code, with up to 2 repair re-prompts. Beat pictures generate as a teacher writes: a batch credit reservation first, a prompt carrying the lesson and a translation gloss (“cone” stopped being ice cream), sanitised inside the builder. Voice: the model writes emotion tags, a deterministic compiler writes the SSML. promptfoo A/B suites guard both.

STACK → Gemini, Ideogram v4, Resemble, promptfoo, OpenTelemetry

SYS-04 THE PLATFORM UNDER IT

~190 API handlers, 43 Prisma models, 73 migrations; Postgres RLS forced on 28 tables by a script that verifies itself. A credit ledger where reserve, commit and refund share one transaction and row lock; idempotency keys; Stripe webhook dedupe. Rotating refresh tokens with reuse detection, TOTP MFA, child logins with no email and PII-free tokens. ~6,600 tests; expand/contract releases.

STACK → Fastify, Prisma, Postgres, Next.js 16, React 19, Stripe, Turborepo, Cloud SQL, Vercel

DOCUMENT	PROJECT	SCALE	DATE	SHEET
Curriculum Vitae — Emile Harmel	Hobo Labs	1:1	2026-09	1 / 2

- CO-FOUNDER & CHIEF TECHNOLOGIST

MonstoryX

2024 – present
- Architected and built the teacher platform end to end (§02): 25-workspace TypeScript monorepo, 7 apps, 13 domains, 1,569 commits in 14 weeks. Pre-alpha with teachers.
- Designed the agent systems – co-author, quest images, voice, the Finale video pipeline – and the reliability, evaluation and billing layers around them.
- Run engineering as a multi-agent team: owner agents hired by the paths a change touches, blind critics, memory as the bus, every binding decision an ADR.
- Also: the studio's animation line (script → keyframes → voice → lip-sync), output public at youtube.com/@monstoryx/shorts.
- LEAD PRODUCT ARCHITECT

XiTrust / Moxis

2024 – 2026
- Frontend re-architecture of an enterprise e-signature platform: Nx monorepo on Next.js 16 / React 19 / TypeScript with a shared design system, and local LLMs turning static forms into conversational workflows.
- FULL-STACK DEVELOPER – HOBO LABS

Nutrons / Awesome Nuts GmbH

2026
- Gamified web app, fixed fee, hard launch: Next.js / TypeScript, Postgres (RLS), QR-scan reward economy and leaderboard, generation and validation of 20,000 unique codes.
- IOS DEVELOPER

Rise World

2021 – 2024
- Key developer on a Gematik app in Germany's national e-health programme – secure, accessible, compliance-audited software used by millions.
- LEAD CREATIVE DEVELOPER & TECHNICAL DIRECTOR

Demodern (agency)

2016 – 2021
- Find My Mazda – led design and build of Mazda Germany's online sales platform; conversion +18%, qualified dealer inquiries +30%. My Mazda – architected the app across all EU markets.
- Super RTL – conversational interface for children at broadcast scale. Nestlé Wagner – Spark AR campaigns.

CORE	TypeScript (strict), Node.js 22, SQL · Next.js 16, React 19, TanStack Query, Tailwind, Radix, next-intl, React Three Fiber
BACKEND & DATA	Fastify 5, Hono, Prisma 6, Postgres 16 (RLS, row locks, idempotency), Stripe, BigQuery billing export, pnpm + Turborepo
AGENTS & AI	Flue agent runtime, Gemini (text & vision), Ollama Cloud, fal.ai (Ideogram v4, Nano Banana 2, Wan 3.0), Resemble TTS, ffmpeg
EVALS & OPS	promptfoo, LLM & vision judges, Vitest, Playwright, MSW, OpenTelemetry, pino, GitHub Actions
INFRA	Cloud Run, Cloud SQL, Cloud Build, Artifact Registry, Docker, Vercel
AGENTIC DEV	Claude Code (subagents, skills, hooks, MCP), owner/critic agent teams, memory-as-bus, ADR-governed decisions
ALSO	Swift / iOS, React Native, Python (FastAPI), Unreal Engine 5, Figma

City College Manchester – HND Multimedia Design · ND Art Foundation with Mathematics · City & Guilds Project Management

Languages – English (mother tongue) · German (Grundkenntnisse, actively improving – I live in Graz)

Repository private – every figure on this sheet is measured from it, and I will screen-share any part of it.

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